

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B. Tech (CBCS) Degree Examinations

December 2024

EC3211-1- DSP PROCESSORS AND ARCHITECTURES

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **DSP uses _____ memory architecture.**
 - A) Von Neumann
 - B) Harvard Architecture with one data memory
 - C) Harvard Architecture with two data memories
 - D) Harvard Architecture with three data memories
2. **Overflow/underflow can be avoided in DSP by using**
 - A) Shifters at the input and output
 - B) By providing guard bits in the accumulator
 - C) Saturation logic
 - D) All of these
3. **A pipelined MAC unit requires _____ clock cycles to compute the sum of N products.**
 - A) N+1
 - B) N
 - C) 2N
 - D) 2N+1
4. **Bit reversed addressing is used for**
 - A) FIR filters
 - B) FFT computation
 - C) IIR filters
 - D) Circular convolution
5. **The central ALU of C54X DSP processors have _____ bit multiplier.**
 - A) 32x32
 - B) 16x16
 - C) 17x17
 - D) 64x64
6. **The number of guard bits provided in accumulator of C54X DSP processors is _____**
 - A) 32
 - B) 16
 - C) 8
 - D) 4
7. **The register which holds the address of the current data memory page is _____.**
 - A) DP
 - B) ARP
 - C) ST0
 - D) ST1
8. **Assuming AR4= 35h what is the contents of AR4 after the execution *AR4-B**
 - A) 34h
 - B) 3Fh
 - C) 31h
 - D) 36h
9. **Using the RPT #k instruction, the maximum no. of times a single instruction can be repeatedly executed is _____.**
 - A) 65535
 - B) 255
 - C) 256
 - D) 65536
10. **Which of the following instruction does not alter the value of AR2?**
 - A) LD*AR2, 20h
 - B) LD *AR2+, 20h
 - C) LD *+AR2(2), 20h
 - D) LD *AR2(2), 20h
11. **The instruction for adding the content of the location pointed by AR2 to that pointed by AR3 and storing the result in A register is _____.**
 - A) ADD A, AR2, AR3
 - B) ADD A, *AR3, *AR4
 - C) ADD A, *AR2, *AR3
 - D) ADD *AR2, *AR3, A

12. The content of the accumulator after the execution of the instruction LD *AR1, A, if the content of the memory pointed by AR1 is 8000H and SXM=1.
 A) 00 0000 8000H B) FF FFFF 8000H
 C) FF 0000 8000 H D) 00 FFFF 8000H
13. HPI has _____ bit data bus and the _____ bit address bus.
 A) 16, 18 B) 18, 16
 C) 8, 16 D) 16, 8
14. Devices of TMS320C54xx family have _____ interrupts.
 A) 32 B) 16
 C) 20 D) 24
15. A P-DSP has four pipeline stages and uses four phase clock. The number of clock cycles required for executing a program with 25 instruction is _____.
 A) 29 B) 28
 C) 25 D) 26
16. _____ interrupts are serviced whenever they occur
 A) Maskable B) Non-maskable
 C) Both of these D) None of these
17. The direct DFT computation of these equations require _____ complex multiplies.
 A) N B) N^2
 C) 2N D) $N(N-1)$
18. Number of butterflies needed for the entire computation in a 64-point FFT.
 A) 65 B) 67
 C) 63 D) 64
19. Encoding an image of a 256 x 256 size requires approximately _____ in the TMS320, with an instruction cycle of 10 ns.
 A) 100 msec B) 150 msec
 C) 140 msec D) 120 msec
20. A decimation filter is used to _____ the sampling rate of the signal.
 A) Increase B) Reduce
 C) keep D) change

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat diagram explain arithmetic and logical unit (ALU) of DSP system.	6	L*2	1	1
	b) Explain different methods available to handle overflow and underflow in MAC unit.	6	L2	1	1
	c) Compare architectural features of TMS320C25 and DSP56000 fixed point digital signal processors.	4	L2	2	1
2.	a) Explain the features of a program sequencer unit of a programmable DSP with a neat block diagram.	6	L2	1	1
	b) Explain status registers ST0 and ST1 of TMS320C54XX processor.	6	L2	2	1
	c) A DSP has a circular buffer with the start and end addresses of the circular buffer are 0210h and 0201h, respectively. What would be the new values of the address pointer of the buffer if, in the course of address computation, it gets updated to (a) 0212h, (b) 01FCh?	4	L3	1	1
3.	a) Give the structure of a 4x4 Braun multiplier, explain its concept.	6	L2	1	1
	b) Identify the addressing modes of the following operands. Assume that the current contents of AR3 to be 400 h, what will be its contents after each of the following TMS320C54XX addressing mode is used? Assume that the contents of AR0 are 40 h. i) *AR3+0 ii) *AR3+ iii) *AR3-0B	6	L2	2	1

- c) Compute the sequence in which the input data should be ordered for a 16 point DIT FFT.

4 L3 1 1

Unit – II

4. a) Describe the operation of the following instructions of TMS320C54xx processor, with an example. 1) MAC 2) RPT 3) MPY 4) MAS
b) Describe Host Port interface and explain its signals.

8 L3 3 1
8 L2 4 1

5. a) By means of a figure explain the pipeline operation of the following sequence of instruction if the initial values of AR1, AR3, A are 84, 81, 1 and the values stored in the memory locations 81, 82, 83, 84 are 2, 3, 4, 6. Also provide the values of registers AR3, AR1, T and accumulator after completion of each cycle.

ADD *AR3+, A

LD *AR1+, T

MPY *AR3+, B

ADD B, A

8 L3 4 1

- b) Write a program to compute the Sum of three product terms given by the equation

$$y(n) = h_0x(n) + h_1x(n-1) + h_2x(n-2)$$

where $x(n)$, $x(n-1)$ and $x(n-2)$ are data samples stored at three successive data memory locations and h_0 , h_1 and h_2 are constants stored at three other successive locations in the data memory. The result $y(n)$ is to be stored in the data memory. Use direct addressing mode to access the data memory.

8 L3 3 1

6. a) Find the contents of the Accumulator A, B and AR3 after the execution of the following instructions. If the contents of the A=1200 H, B=1800H, AR3=0100H and 1500 H is stored initially in 0100 H.

(i) ADD *AR3+, 14, A (ii) ADD A, -8, B.

8 L3 3 1

- b) Draw the logical block diagram of timer circuit and explain the operation of on chip timer of TMS320C54xx DSP.

8 L2 4 1

Unit – III

7. a) Explain how the bit-reversed index generation can be done in 8 point FFT. Also write a TMS320C54XX program for 8 point DIT FFT bit reversed index generation.

6 L3 5 1

- b) With a neat diagram explain Synchronous Serial Interface of TMS320C54XX processor.

6 L2 5 1

- c) Write an assembly language program for TMS320C54XX processors to multiply two Q15 numbers to produce Q15 number result.

4 L3 5 1

8. a) Derive the optimum scaling factor for the DIT FFT butterfly structure.

6 L3 5 1

- b) What do you understand by a DSP based biotelemetry receiver? Explain with a block diagram DSP-Based Biotelemetry Receiver system.

6 L2 5 1

- c) Determine the value of each of the following 16-bit numbers represented using the given Q-notations: 1) 4400h as Q15 number 2) 4400h as a Q7 number 3) 4000h as a Q15 number 4) 4000h as a Q7 number.

4 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
